Light-Emitting Diode Junction-Temperature Sensing using Differential Voltage/Current Measurements

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***Abstract*—Given the temperature dependence of various aspects of light-emitting diode (LED) performance, LED temperature sensing is becoming increasingly important in solid-state lighting applications. This paper presents an electrical technique for junction-temperature sensing based on the measurement of the forward voltage and current of an LED at two bias points. This technique is inspired by techniques commonly used in temperature sensors based on bipolar transistors. While it leads to higher temperature errors than existing electrical techniques, which use the linear relationship between voltage and temperature at a fixed current, the proposed technique has the potential to significantly reduce calibration costs, as it requires calibration at only one temperature instead of two for existing techniques. Measurements of commercial high-power LEDs show that temperature errors can be reduced by using differential measurements around a fixed voltage-bias point instead of the more commonly used fixed current-bias point.**

I. Introduction

High-power light-emitting diodes (LEDs) have become an important competitor for traditional light sources for various applications, such as home and office lighting, automotive lights and traffic lights. In spite of their higher efficiency, high-power LEDs dissipate a considerable amount of heat. The associated increase of their junction temperature gives rise to various problems. The first of these is a shorter lifetime [1]. As a result, LED manufacturers define derating specifications, i.e. the maximum current is reduced with rising temperature. Secondly, the light output decreases and finally, the chromaticity changes with temperature [2]. These effects can generally be compensated for, but the junction temperature must be known in order to do that.

There are many examples of LED drivers that utilize a thermistor placed next to the LED package for temperature feedback. The junction temperature is then estimated based on the thermistor temperature [3]. Various other methods are based on the LED's radiated spectrum [4], micro-Raman-spectroscopy [5] or liquid crystal thermography [6]. These methods all make use of an external sensor, adding to the cost and uncertainty of the measurement.

The first technique for measuring an LED's junction temperature electrically dates back to 1977 [7] and has been evaluated a number of times since then [8]–[10]. It is based on the linear relationship between forward voltage and junction temperature at constant-current bias. Although it does not require an extra sensor, this technique requires a calibration at two temperatures, making it still quite expensive to implement on a large scale.

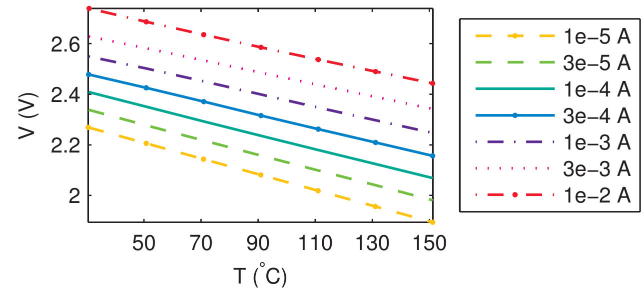

Fig. 1. Measured linear relationship between voltage and temperature for different bias currents (LED type A).

The techniques presented in this paper are based on the differential voltage/current measurement techniques extensively used in bipolar transistor-based thermometers and, in principle, require calibration at one temperature only. In this paper, we investigate experimentally what performance can be obtained when applying these techniques to LEDs.

II. Theory

A. LED Model

To calculate the temperature of an LED from its electrical characteristics, the mathematical relations between temperature (T), voltage (V) and current (I) must first be known. The current in an ideal diode is given by the Shockley equation [11]:

$$I = I_S(e^{\frac{qV}{nkT}} - 1), \tag{1}$$

where I_S, q and k are the temperature dependent saturation current, elementary charge and Boltzmann's constant, respectively. Any non-idealities are modeled by the non-ideality factor, n ($n \geq 1$). Furthermore, at typical forward bias voltages, i.e. $V \gg kT/q$, the -1 term can be neglected.

B. One-point technique

Existing techniques use the linear relationship between forward voltage and junction temperature at a constant current (Fig. 1). If this current is smaller than the current used during normal operation of the LED, it can be applied in a time-multiplexed manner, so that the normal operation current is periodically briefly interrupted for the measurement. This

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relationship is typically derived experimentally using calibrations at two temperatures [10]. This yields a temperature dependence of the form

$$T = \alpha + \beta V, \tag{2}$$

where α and β follow from two voltages V_{cal1} and V_{cal2} measured at known temperatures T_{cal1} and T_{cal2}, respectively:

$$\beta = \frac{T_{cal2} - T_{cal1}}{V_{cal2} - V_{cal1}}, \alpha = T_{cal1} - V_{cal1}\beta. \tag{3}$$

This technique is reported to yield accuracies of ±3°C [12].

C. Two-point technique

The two-point technique combines two measurements to calculate the junction temperature of a diode [13]. The junction is consecutively biased with two currents, I_1 and $I_2 = pI_1$. The resulting voltage difference is PTAT (proportional to absolute temperature):

$$\Delta V = V_{I_2} - V_{I_1} = \frac{nkT}{q} \ln \frac{I_2}{I_S} - \frac{nkT}{q} \ln \frac{I_1}{I_S} = \frac{nkT}{q} \ln p. \tag{4}$$

This expression is not dependent on I_S. As a result, this technique has the advantage of requiring only calibration at one temperature, as n is the only unknown variable.

III. Measurements

The goal of the measurements presented in this paper is to determine the accuracy with which the junction temperature of an LED can be determined from its electrical I-V characteristics using the above-mentioned techniques. That is, the temperature error T_{err} (the difference between the measured temperature and the actual temperature) must be determined.

A. Methodology

The different techniques have been evaluated by applying them to various LEDs. To be able to do this, each LED was first characterized over a broad range of currents, voltages and temperatures, resulting in a set of measurement data for each LED. The measurement techniques were then applied to these datasets using Matlab. Four LED types (labeled A–D) have been characterized. These are all commercially available, high-power (1W, 350mA@3V) white LEDs with a single die of 1mm^2 and different color temperatures.

B. Measurement setup

A automated measurement setup (Fig. 2 and 3) has been built to accurately measure the relevant quantities at the required temperatures and bias points. The temperature is controlled using a Vötsch VT7004 remote-controlled thermal chamber, while temperature is measured using Pt-100 thermistors embedded in an aluminum block, placed inside the chamber, on which the LEDs are mounted. The LEDs are biased with a current or voltage that is constant over temperature with a Keithley 2400 or Yokogawa GS200 source, respectively. Current and voltage measurements are performed using a Keithley 2001 and 2002 DMM, respectively. Up to

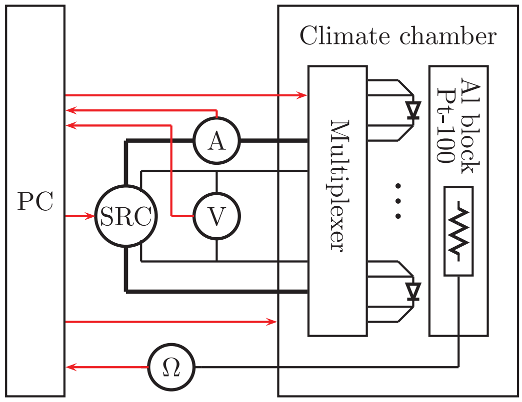

Fig. 2. Diagram of the measurement setup used.

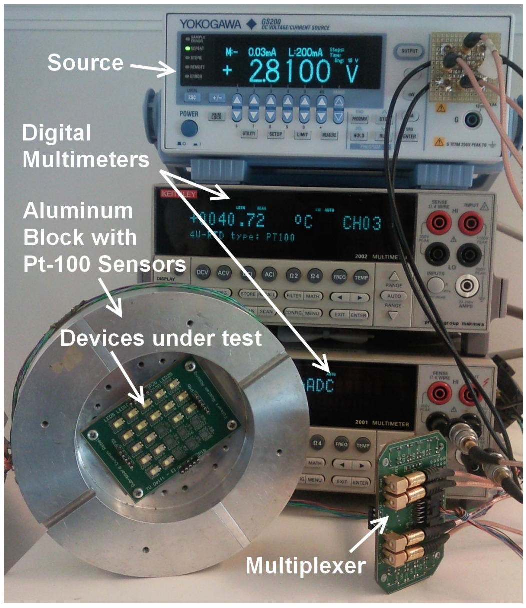

Fig. 3. Photograph of measurement setup used.

25 LEDs can be measured simultaneously using a custom-built relay-based remote controlled multiplexer system. A four-wire measurement is used to measure voltages. Self-heating of the LEDs during measurement is prevented by using pulsed measurements at high currents.

IV. Experimental results

A. One-point technique

The existing one-point technique is applied to enable a fair comparison with the other investigated techniques. Fig. 4 shows T_{err} obtained using this technique as a function of the bias current used, for a sample of LED type A measured at various temperatures. The coefficients α and β in (2) were derived from a calibration at 70°C and 110°C. The peak-to-peak errors across the temperature range from 50°C–130°C are plotted for all four LED types in the top-left graph of Fig. 5, again as a function of the bias current. In case the error is always positive or negative across the full temperature range,

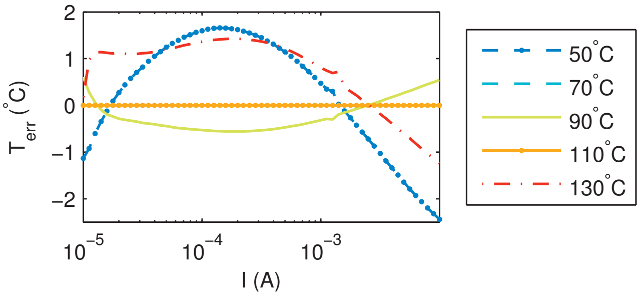

Fig. 4. Measured T_{err} vs. bias current for different temperatures (for LED A).

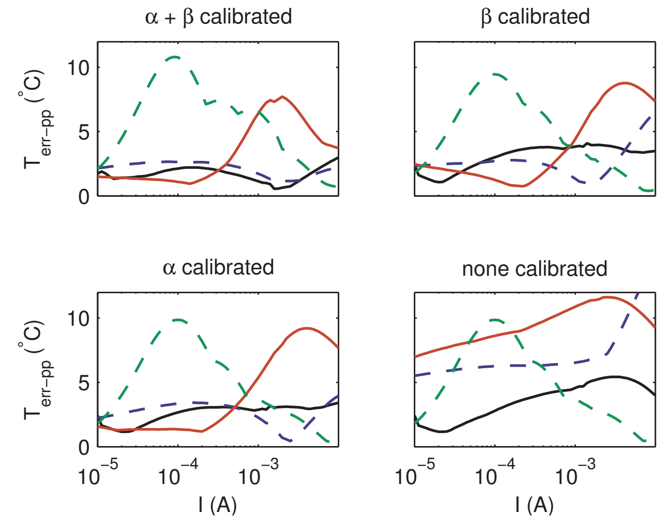

Fig. 5. Measured peak-to-peak error T_{err-pp} using one-point techniques with current biasing. Legend of Fig. 6 applies.

the peak-to-zero error is shown instead, since the peak-to-peak error is then less representative for the worst-case error. For each LED, there are a number of bias currents for which T_{err-pp} is well below the ±3°C mentioned in literature.

To form a better comparison with the proposed techniques, batch-calibration is used to reduce the need for calibration. Three different batch calibration schemes are tested: with an individually calibrated α, with an individually calibrated β and with both coefficients batch-calibrated. For batch-calibration, the average coeffient for all measured samples is used. The results are shown in the other plots of Fig. 5. Individually calibrating only α or β has quite a small influence on the temperature errors, whereas batch calibrating both coefficients worsens the results for most of the range.

B. Two-point technique

The two-point technique is based on the PTAT charachteristic of ΔV at known currents I_1 and I_2, as described in Eq. (4). Because of the unknown n of an LED, a calibration at one temperature T_{cal} is necessary. From this, n is calculated, which is then assumed to be independent of temperature. Fig. 6 shows how this technique performs for different bias currents.

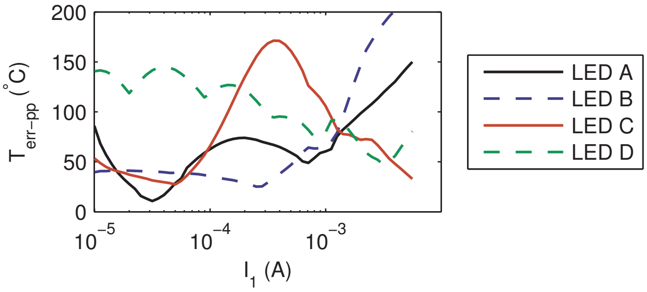

Fig. 6. Measured peak-to-peak error T_{err-pp} using two-point technique and current biasing for four LED types.

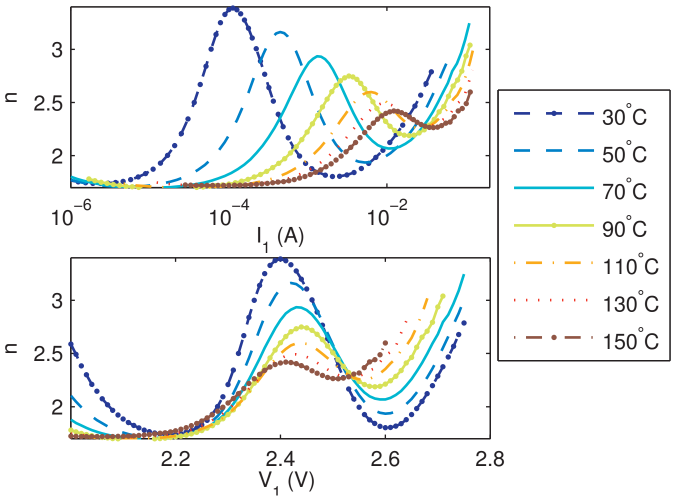

Fig. 7. Non-ideality factor of LED C with different current biasing (top) and voltage biasing (bottom) ($I_2 = 1.78 I_1$).

Part of the poor performance of some LED types is caused by parasitic series resistance. A technique to cancel series resistance [14] was applied and was found to yield an improvement at high currents. The best results however, occuring at low currents, were not affected so this technique does not improve the overall results.

To gain insight in the causes for the other variations, n was investigated. Fig. 7 shows that the non-ideality factor is not constant over temperature, causing large measurement errors. Interestingly, when n is plotted vs. the voltage instead of the current, the bumps in the curves line up, resulting in much less variation over temperature. This implies that, at least for LED types B and C, better accuracy can be expected when using constant-voltage rather than constant-current bias points. This is confirmed by the peak-to-peak errors shown in Fig. 8.

C. Statistical comparison

To analyze the performance for multiple LEDs, a number of LEDs (12 for voltage-biased measurements, 25 for current-biased measurements) is compared over a temperature range of 50–130°C.

For each LED, each technique has an ideal bias point. Deviation from this point causes a larger T_{err}. Measured errors

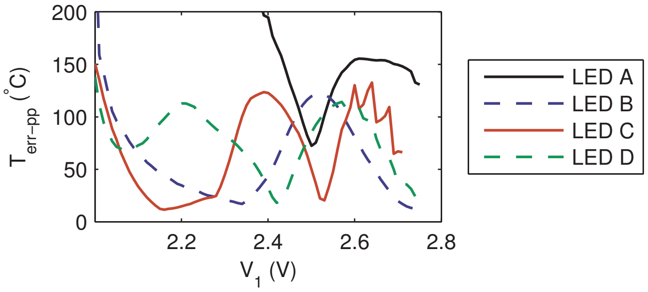

Fig. 8. Measured peak-to-peak error T_{err-pp} using two-point technique and voltage biasing for four LED types.

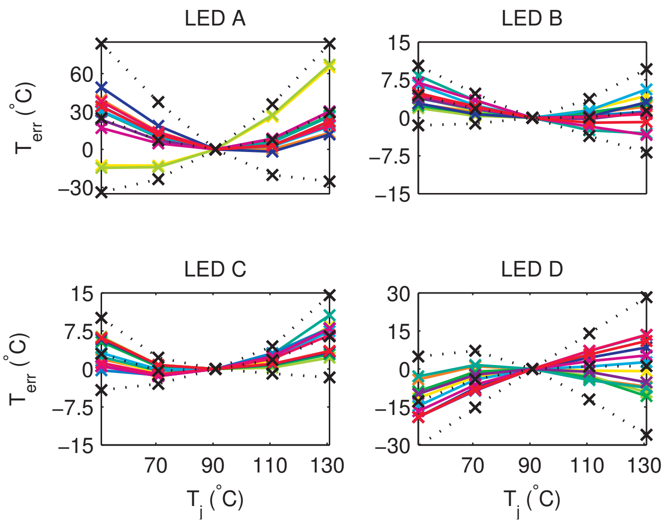

Fig. 9. Family plot of T_{err} for 12 LEDs using two-point technique and best voltage biasing point for four LED types. Dotted lines are mean and $\pm 3\sigma$ of distribution.

for the two-point technique for all four LED types, each biased at its optimal voltage-bias point, are shown in Fig. 9. The peak-to-peak errors for all techniques are listed in Table I.

V. Discussion and Conclusion

When comparing the measurement techniques, the one-point technique with two-temperature calibration stands out and performs as described in literature. However, when batch calibration is applied to reduce the number of calibration

Tech.	Cal.	Bias.	A	B	C	D
1-pt	$\alpha + \beta$	I	2.34	1.96	1.52	0.99
1-pt	α	I	6.79	5.33	3.62	4.78
1-pt	β	I	7.32	17.6	3.22	3.89
1-pt	none	I	20.3	23.4	8.61	9.69
1-pt	$\alpha + \beta$	V	2.76	2.92	3.05	2.95
2-pt	n	I	79.0	41.3	26.0	19.9
2-pt	n	V	81.4	11.7	11.0	32.6

TABLE I
Comparison of peak-to-peak T_{err}(°C) for various techniques and four types of LEDs. Individually calibrated parameters are given in 'cal.' column.

points, T_{err} increases significantly.

Results for the two-point technique are very dependent on the type of LED used. Nevertheless, even for the best LED types, they are not as good as for the one-point technique. The reason this technique does not seem to work well on LEDs can be seen when looking at n. With a highly temperature dependent n, LEDs deviate significantly from the ideal diode characteristic, upon which this technique is based.

When using constant-voltage biasing instead of constant-current biasing, some LEDs show significant improvements. This can be seen as a technique to partly circumvent the non-idealities in the diode characteristic.

A factor that adds uncertainty to the measurements is drift in the device characteristics. Although this effect is limited within one set of measurements (performed over one cycle of temperatures), two successive sets of measurements show differences leading to temperature errors of several degrees. This may be a burn-in effect. The influence of this effect on all techniques is a topic for further research.

Acknowledgment

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References

[1] Philips Lumileds, "Luxeon Rebel RD07 reliability data," Nov. 2007.
[2] Nichia Corporation, "NCSL119-H3 datasheet."
[3] N. Naredran, Y. Gu, and R. Hosseinzadeh, "Estimating the junction temperature of high-flux white LEDs," in *Proceedings of SPIE 5366*, S. Stockman, H. Yao, and E. Schubert, Eds. International Society of Optical Engineers, 2004, pp. 158–160.
[4] Z. Vaitonis, P. Vitta, and A. Zukauskas, "Measurement of the junction temperature in high-power light-emitting diodes from the high-energy wing of the electroluminescence band," *Journal of Applied Physics*, vol. 103, no. 9, May 2008.
[5] Y. Wang, *et al.*, "Determination of junction temperature of GaN-based light emitting diodes by electroluminescence and micro-raman spectroscopy," in *CS MANTECH Conference*, 2009.
[6] C. C. Lee and J. Park, "Temperature measurement of visible light-emitting diodes using nematic liquid crystal thermography with laser illumination," *IEEE Photonics Technology Letters*, vol. 16, pp. 1706–1708, 2004.
[7] B. Griffing and S. Shivashankar, "Use of light-emitting diodes as temperature sensor," *Rev. Sci. Instrum.*, vol. 48, no. 9, pp. 1225–1226, September 1977.
[8] Y. Archaya and P. Vyavahare, "Study on the temperature sensing capability of a light-emitting diode," *Review of Scientific Instruments*, vol. 68, no. 12, December 1997.
[9] Y. Yang, *et al.*, "Junction temperature measurement of light-emitting diodes by voltage-temperature relation method," in *Conference on Lasers and Electro-Optics - Pacific Rim*, 2007.
[10] A. Keppens, *et al.*, "High power light-emitting diode junction temperature determination from current-voltage characteristics," *Journal of Applied Physics*, vol. 104, 2008.
[11] D. A. Neamen, *Semiconductor Physics and Devices: Basic Principles*, 3rd ed. McGraw-Hill, 2003.
[12] Y. Xi, *et al.*, "Junction and carrier temperature measurements in deep-ultraviolet light-emitting diodes using three different methods," *Applied Physics Letters*, vol. 86, no. 3, January 2005.
[13] T. Verster, "P-n junction as an ultraliear calculable thermometer," *Electronics Letters*, vol. 4, no. 9, pp. 175–176, May 1968.
[14] M. Pertijs and J. Huijsing, *Precision Temperature Sensors in CMOS Technology*. Springer, 2006.